LAB ASSIGNMENT – 9
DATE: 06/04/2026

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RollNo: UG/02/BTCSEAIML/2023/075
RegNO: AU/2023/0010993
Section :'E'

COLLECTION 1: **members**

```
{
  member_id: Number,
  name: String,
  plan: String,
  city: String,
  fee: Number
}
```

Table

member_id	name	plan	city	fee
1	Aarav	Gold	Delhi	5000
2	Meera	Silver	Mumbai	3000
3	Rohan	Gold	Bangalore	5500
4	Sneha	Platinum	Delhi	7000
5	Kunal	Silver	Chennai	3200

COLLECTION 2: **sessions**

{

session_id: Number,

member_id: Number,

trainer: String,

duration: Number,

calories: Number

}

Table

session_id	member_id	trainer	duration	calories
101	1	Raj	60	400
102	2	Simran	45	300
103	3	Raj	70	500
104	4	Aman	80	600
105	5	Raj	50	350

```
test> use GymDB_075
switched to db GymDB_075
GymDB_075> db.members_075.insertMany([{member_id:1,name:"Aarav",plan:"Gold",city:"De
lhi",fee:5000},{member_id:2,name:"Meera",plan:"Silver",city:"Mumbai",fee:3000},{memb
member_id:3,name:"Rohan",plan:"Gold",city:"Bangalore",fee:5500},{member_id:4,name:"S
neha",plan:"Platinum",city:"Delhi",fee:7000},{member_id:5,name:"Kunal",plan:"Silver"
,city:"Chennai",fee:3200}])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69d3547ea8252236611e2621'),
    '1': ObjectId('69d3547ea8252236611e2622'),
    '2': ObjectId('69d3547ea8252236611e2623'),
    '3': ObjectId('69d3547ea8252236611e2624'),
    '4': ObjectId('69d3547ea8252236611e2625')
  }
}
GymDB_075> db.sessions_075.insertMany([{session_id:101,member_id:1,trainer:"Raj",dur
ation:60,calories:400},{session_id:102,member_id:2,trainer:"Simran",duration:45,calo
calories:300},{session_id:103,member_id:3,trainer:"Raj",duration:70,calories:500},{s
session_id:104,member_id:4,trainer:"Aman",duration:80,calories:600},{session_id:105,
member_id:5,trainer:"Raj",duration:50,calories:350}])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69d3549ea8252236611e2626'),
    '1': ObjectId('69d3549ea8252236611e2627'),
    '2': ObjectId('69d3549ea8252236611e2628'),
    '3': ObjectId('69d3549ea8252236611e2629'),
    '4': ObjectId('69d3549ea8252236611e262a')
  }
}
```

QUESTIONS

Q1. Display all members

```
GymDB_075> db.members_075.find()
[
  {
    _id: ObjectId('69d3547ea8252236611e2621'),
    member_id: 1,
    name: 'Aarav',
    plan: 'Gold',
    city: 'Delhi',
    fee: 5000
  },
  {
    _id: ObjectId('69d3547ea8252236611e2622'),
    member_id: 2,
    name: 'Meera',
    plan: 'Silver',
    city: 'Mumbai',
    fee: 3000
  },
  {
    _id: ObjectId('69d3547ea8252236611e2623'),
    member_id: 3,
    name: 'Rohan',
    plan: 'Gold',
    city: 'Bangalore',
    fee: 5500
  },
  {
    _id: ObjectId('69d3547ea8252236611e2624'),
    member_id: 4,
    name: 'Sneha',
    plan: 'Platinum',
    city: 'Delhi',
    fee: 7000
  },
  {
    _id: ObjectId('69d3547ea8252236611e2625'),
    member_id: 5,
    name: 'Kunal',
    plan: 'Silver',
    city: 'Chennai',
    fee: 3200
```

Q2. Show name and plan only

```
GymDB_075> db.members_075.find({}, { _id: 0, name: 1, plan: 1 })
[
  { name: 'Aarav', plan: 'Gold' },
  { name: 'Meera', plan: 'Silver' },
  { name: 'Rohan', plan: 'Gold' },
  { name: 'Sneha', plan: 'Platinum' },
  { name: 'Kunal', plan: 'Silver' }
]
```

Q3. Find members from Delhi

```
GymDB_075> db.members_075.find({ city: "Delhi" })
[
  { name: 'Aarav', plan: 'Gold' },
    _id: ObjectId('69d3547ea8252236611e2621'),
    member_id: 1,, plan: 'Gold' },
    name: 'Aarav', plan: 'Platinum' },
    plan: 'Gold',, plan: 'Silver' }
    city: 'Delhi',
    fee: 5000
  },
  {
    _id: ObjectId('69d3547ea8252236611e2624'),
    member_id: 4,
    name: 'Sneha',
    plan: 'Platinum',
    city: 'Delhi',
    fee: 7000
  }
]
```

Q4. Find sessions with duration > 60

```
GymDB_075> db.sessions_075.find({ duration: { $gt: 60 } })
[
  {
    _id: ObjectId('69d3549ea8252236611e2628'),
    session_id: 103,
    member_id: 3,
    trainer: 'Raj',
    duration: 70,
    calories: 500
  },
  {
    _id: ObjectId('69d3549ea8252236611e2629'),
    session_id: 104,
    member_id: 4,
    trainer: 'Aman',
    duration: 80,
    calories: 600
  }
]
```

Q5. Sort members by fee descending

```
GymDB_075> db.members_075.find().sort({ fee: -1 })
[
  {
    _id: ObjectId('69d3547ea8252236611e2624'),
    member_id: 4,
    name: 'Sneha',
    plan: 'Platinum',
    city: 'Delhi',
    fee: 7000
  },
  {
    _id: ObjectId('69d3547ea8252236611e2623'),
    member_id: 3,
    name: 'Rohan',
    plan: 'Gold',
    city: 'Bangalore',
    fee: 5500
  },
  {
    _id: ObjectId('69d3547ea8252236611e2621'),
    member_id: 1,
    name: 'Aarav',
    plan: 'Gold',
    city: 'Delhi',
    fee: 5000
  },
  {
    _id: ObjectId('69d3547ea8252236611e2625'),
    member_id: 5,
    name: 'Kunal',
    plan: 'Silver',
    city: 'Chennai',
    fee: 3200
  },
  {
    _id: ObjectId('69d3547ea8252236611e2622'),
    member_id: 2,
    name: 'Meera',
    plan: 'Silver',
    city: 'Mumbai',
    fee: 3000
  }
]
```

Q6. Update fee +500 for Gold members

```
GymDB_075> db.members_075.updateMany( { plan: "Gold" }, { $inc: { fee: 500 } } )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
```

Q7. Delete members with fee < 3500

```
GymDB_075> db.members_075.deleteMany({fee:{$lt:3500}})
{ acknowledged: true, deletedCount: 2 }
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
```

Q8. Insert a new member

```
  acknowledged: true,
  insertedId: ObjectId('69d357e3a8252236611e262b')
} matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
```

Q9. Average fee per plan

```
GymDB_075> db.members_075.aggregate([{$group:{_id:"$plan",avg_fee:{$avg:"$fee"}}}])
[ { _id: 'Platinum', avg_fee: 7000 }, { _id: 'Gold', avg_fee: 5475 } ]
  acknowledged: true,
  insertedId: ObjectId('69d35812a8252236611e262c')
} matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
```

Q10. Total calories burned per trainer

```
GymDB_075> db.sessions_075.aggregate([{$group:{_id:"$trainer",total_calories:{$sum:"$calories"}}}])
[
  { _id: 'Aman', total_calories: 600 },
  { _id: 'Raj', total_calories: 1250 },
  { _id: 'Simran', total_calories: 300 }
]
```

Q11. Maximum fee per city

```
GymDB_075> db.members_075.aggregate([{$group:{_id:"$city",max_fee:{$max:"$fee"}}}])
[
  { _id: 'Bangalore', max_fee: 6000 },
  { _id: 'Delhi', max_fee: 7000 },
  { _id: 'Kolkata', max_fee: 5200 }
]
```

Q12. Count members per plan

```
GymDB_075> db.members_075.aggregate([{$group:{_id:"$plan",total_members:{$sum:1}}}])
[
  { _id: 'Platinum', total_members: 1 },
  { _id: 'Gold', total_members: 4 }
]
```

Q13. Trainers with avg calories > 400

```
GymDB_075> db.sessions_075.aggregate([{$group:{_id:"$trainer",avg_calories:{$avg:"$calories"}}},{$match:{avg_calories:{$gt:400}}}])
[
  { _id: 'Raj', avg_calories: 416.6666666666667 },
  { _id: 'Aman', avg_calories: 600 }
]
```

Q14. Total duration per member_id

```
GymDB_075> db.sessions_075.aggregate([{$group:{_id:"$member_id",total_duration:{$sum:"$duration"}}}])
GymDB_075>
  { _id: 5, total_duration: 50 },
  { _id: 2, total_duration: 45 },
  { _id: 1, total_duration: 60 },
  { _id: 3, total_duration: 70 },
  { _id: 4, total_duration: 80 }
]
```

Q15. Join members with sessions

```
GymDB_075> db.members_075.aggregate([{$lookup:{from:"sessions_075",localField:"member_id",foreignField:"member_id",as:"session_details"}}])
[ { _id: 5, total_duration: 50 },
  { _id: 2, total_duration: 45 },
  { _id: 1, total_duration: 60 },
  { _id: 3, total_duration: 70 },
  { _id: 4, total_duration: 80 }
]
GymDB_075> db.members_075.aggregate([{$lookup:{from:"sessions_075",localField:"member_id",foreignField:"member_id",as:"session_details"}}])
[ { _id: 5, total_duration: 50 },
  { _id: 2, total_duration: 45 },
  { _id: ObjectId('69d3547ea8252236611e2621'),
    member_id: 1,_duration: 70 },
    name: 'Aarav',duration: 80 }
    plan: 'Gold',
    city: 'Delhi',
    fee: 5500,
    session_details: [
      {
        _id: ObjectId('69d3549ea8252236611e2626'),
        session_id: 101,
        member_id: 1,
        trainer: 'Raj',
        duration: 60,
        calories: 400
      }
    ]
  },
  {
    _id: ObjectId('69d3547ea8252236611e2623'),
    member_id: 3,
    name: 'Rohan',
    plan: 'Gold',
    city: 'Bangalore',
    fee: 6000,
    session_details: [
      {
        _id: ObjectId('69d3549ea8252236611e2628'),
        session_id: 103,
        member_id: 3,
        trainer: 'Raj',
        duration: 70,
        calories: 500
      }
    ]
  },
```

```
  {
    _id: ObjectId('69d3547ea8252236611e2624'),
    member_id: 4,
    name: 'Sneha',
    plan: 'Platinum',
    city: 'Delhi',
    fee: 7000,
    session_details: [
      {
        _id: ObjectId('69d3549ea8252236611e2629'),
        session_id: 104,
        member_id: 4,
        trainer: 'Aman',
        duration: 80,
        calories: 600
      }
    ]
  },
  {
    _id: ObjectId('69d357e3a8252236611e262b'),
    member_id: 6,
    name: 'Ishita',
    plan: 'Gold',
    city: 'Kolkata',
    fee: 5200,
    session_details: []
  },
  {
    _id: ObjectId('69d35812a8252236611e262c'),
    member_id: 6,
    name: 'Ishita',
    plan: 'Gold',
    city: 'Kolkata',
    fee: 5200,
    session_details: []
  }
]
```

Q16. Show members who have sessions

```
GymDB_075> db.members_075.aggregate([{$lookup:{from:"sessions_075",localField:"member_id",foreignField:"member_id",as:"session_details"}},{$match:{session_details:{$ne:[]}}
}])
[
  {
    _id: ObjectId('69d3547ea8252236611e2621'),
    member_id: 1,
    name: 'Aarav',
    plan: 'Gold',
    city: 'Delhi',
    fee: 5500,
    session_details: [
      {
        _id: ObjectId('69d3549ea8252236611e2626'),
        session_id: 101,
        member_id: 1,
        trainer: 'Raj',
        duration: 60,
        calories: 400
      }
    ]
  },
  {
    _id: ObjectId('69d3547ea8252236611e2623'),
    member_id: 3,
    name: 'Rohan',
    plan: 'Gold',
    city: 'Bangalore',
    fee: 6000,
    session_details: [
      {
        _id: ObjectId('69d3549ea8252236611e2628'),
        session_id: 103,
        member_id: 3,
        trainer: 'Raj',
        duration: 70,
        calories: 500
      }
    ]
  },
```

Q17. Create index on city

```
GymDB_075> db.members_075.createIndex({city:1})
GymDB_075>
city_1
```

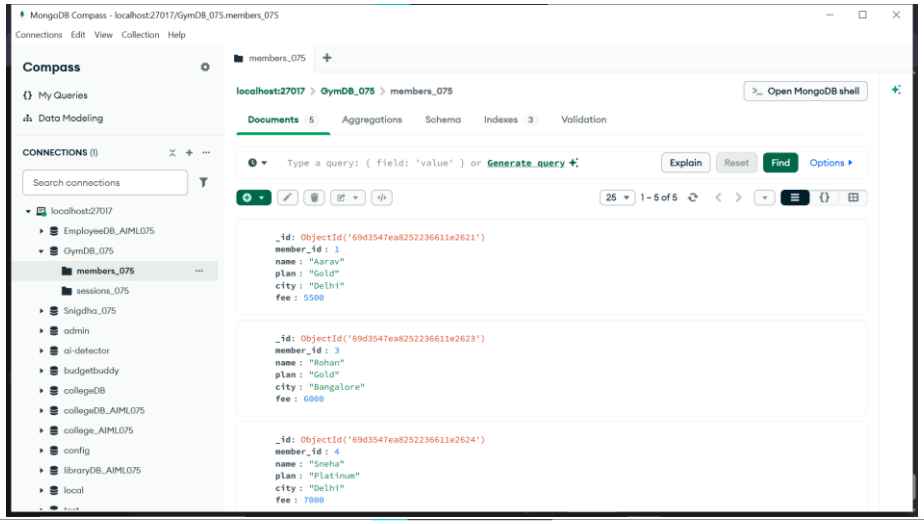
Q19. Create compound index on plan and fee

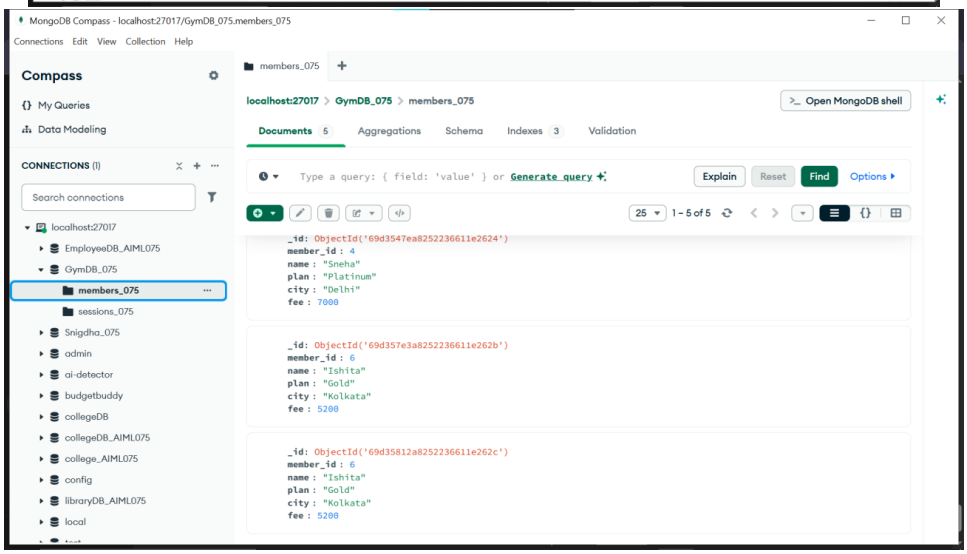
```
GymDB_075> db.members_075.createIndex({plan:1,fee:-1})
plan_1_fee_-1
```

Q20. Display 2nd and 3rd highest fee members

```
GymDB_075> db.members_075.find().sort({fee:-1}).skip(1).limit(2)
[lan_1_fee_-1
  {
    _id: ObjectId('69d3547ea8252236611e2623'),
    member_id: 3,
    name: 'Rohan',
    plan: 'Gold',
    city: 'Bangalore',
    fee: 6000
  },
  {
    _id: ObjectId('69d3547ea8252236611e2621'),
    member_id: 1,
    name: 'Aarav',
    plan: 'Gold',
    city: 'Delhi',
    fee: 5500
  }
]
```

MONGODB COMPASS


MongoDB Compass - localhost:27017/GymDB_075.members_075
Connections Edit View Collection Help
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
localhost:27017
EmployeeDB_AIML075
GymDB_075
members_075
sessions_075
Snigdha_075
admin
ai-detector
budgetbuddy
collegeDB
collegeDB_AIML075
college_AIML075
config
libraryDB_AIML075
local
members_075
localhost:27017 > GymDB_075 > members_075
Open MongoDB shell
Documents 5
Aggregations
Schema
Indexes 3
Validation
Type a query: { field: 'value' } or Generate query
Explain
Reset
Find
Options
25
1 - 5 of 5
_id: ObjectId('69d3547ea8252236611e2621')
member_id : 1
name : "Aarav"
plan : "Gold"
city : "Delhi"
fee : 5500
_id: ObjectId('69d3547ea8252236611e2623')
member_id : 3
name : "Rohan"
plan : "Gold"
city : "Bangalore"
fee : 6000
_id: ObjectId('69d3547ea8252236611e2624')
member_id : 4
name : "Sneha"
plan : "Platinum"
city : "Delhi"
fee : 7000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college_AIML075
config
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localhost:27017 > GymDB_075 > members_075
Open MongoDB shell
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Aggregations
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Type a query: { field: 'value' } or Generate query
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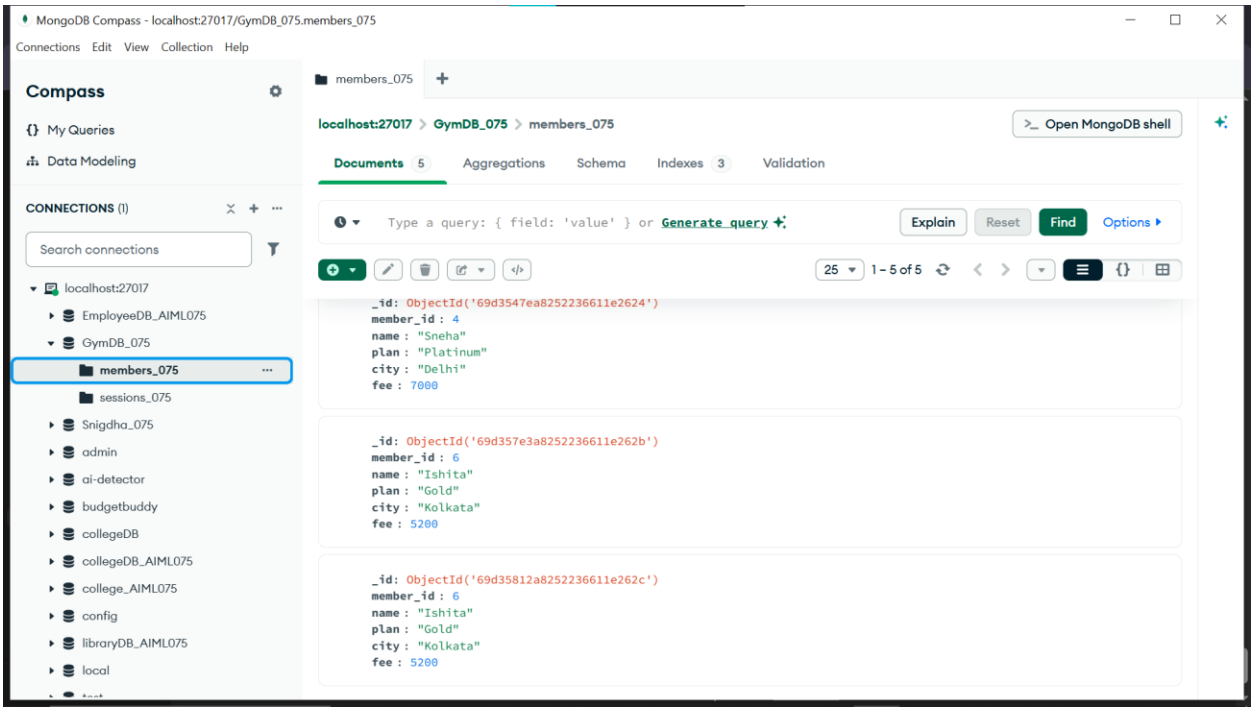
25
1 - 5 of 5
_id: ObjectId('69d3547ea8252236611e2624')
member_id : 4
name : "Sneha"
plan : "Platinum"
city : "Delhi"
fee : 7000
_id: ObjectId('69d357e3a8252236611e262b')
member_id : 6
name : "Ishita"
plan : "Gold"
city : "Kolkata"
fee : 5200
_id: ObjectId('69d35812a8252236611e262c')
member_id : 6
name : "Ishita"
plan : "Gold"
city : "Kolkata"
fee : 5200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_id: ObjectId('69d3547ea8252236611e2624')
member_id : 4
name : "Sneha"
plan : "Platinum"
city : "Delhi"
fee : 7000
_id: ObjectId('69d357e3a8252236611e262b')
member_id : 6
name : "Ishita"
plan : "Gold"
city : "Kolkata"
fee : 5200
_id: ObjectId('69d35812a8252236611e262c')
member_id : 6
name : "Ishita"
plan : "Gold"
city : "Kolkata"
fee : 5200